# When Is Control-Aware Communication Locally Decidable? Information-Structure Limits in Distributed Formation Control

[VERIFIED AUTHOR LIST REQUIRED]

***Abstract*—Control-aware communication assigns resources according to an update's closed-loop consequence, but a distributed sender may lack the information needed to determine it. Exact value reconstruction is sufficient but not necessary for an action decision. We study finite-dimensional sampled linear systems in which a communication action adds a fixed response to a finite-horizon performance output, making its quadratic-cost benefit affine in an augmented closed-loop state. We distinguish reconstructing absolute values, reconstructing relative values, and reproducing an optimal action on an information fiber. Established functional-observability geometry gives the absolute and pairwise row-space tests. A set-valued common-maximizer criterion then directly characterizes fiberwise decidability, with a closed form and decision margin on Euclidean fibers. For threshold-straddling binary fibers we retain exact local minimax deterministic and randomized regret. The minimum noiseless linear-statistic dimensions for all absolute and all relative values differ by at most one; equality is lost precisely through a common hidden offset. These dimensions remain distinct from coalition ownership and acquisition cost.**

**We instantiate the framework on sampled formation controllers with delayed and lossy communication. A predeclared 72-cell deterministic registry covers 5, 7, and 9 agents, three connected graphs, two pinning patterns, two horizons, and two delays. All 1320 unit-response functionals fail exact sender-history row-space identifiability. Across its 448 sender/configuration cells, all 360 nontrivial multi-action sets save exactly one relative-value dimension, but none is fully relative-identifiable. In the frozen five-UAV instance, all ten controller-relevant action types admit independently replayed, unsaturated, dynamically reachable sender-indistinguishable history pairs with opposite binary decisions. The results delimit exact local communication decisions without ruling out prior-specific Bayesian or nonlinear policies.**

***Index Terms*—Networked control systems, distributed control, information structures, functional observability, value of information, event-triggered control, multi-agent formation.**

## I. Introduction

Communication policies for networked control are increasingly asked to send the packet that matters most to the control objective, rather than simply the oldest packet or the packet with the largest state error. Event-triggered control formalizes state- or error-dependent transmissions [1]–[3]; age of information (AoI) formalizes freshness [4], [5]; and value-of-information (VoI) formulations attach a control consequence to an update [6]. These approaches answer different questions. AoI describes how stale a record is, while a finite-horizon control value asks how the closed loop will change if that record is refreshed.

The latter question is meaningful only relative to an information structure. In a distributed formation, the sender knows its own physical and controller state and its transmission history. It need not know the packet currently held by a receiver, the receiver's other stale memories, or remote states that shape the no-action closed-loop trajectory. Acknowledgments can reduce one part of this uncertainty, and ACK-free packet-memory beliefs can be calibrated under a known channel law. Neither fact implies that the total control-action value is locally computable: the cross term between the action response and the global no-action response may remain hidden.

This paper therefore asks questions prior to scheduler design:

> Under what information structures are absolute or relative fixed-response action values identifiable, when is an optimal action nevertheless constant on an information fiber, and what supplementary information recovers the missing values?

The question connects decentralized information structures [7]–[9] with functional observability [10], [11]. We do not claim the row-space tests themselves as new. Rather, we expose the exact communication-action value as the target functional and separate three objects: absolute-value reconstruction, relative-value reconstruction, and fiberwise optimal-action selection. Absolute identifiability implies relative identifiability, which is sufficient for exact ranking, but neither converse holds in general. A fiberwise argmax may be decidable even when relative values are not exactly reconstructible. We then separate missing-statistic dimension from network acquisition cost and close the statements on an implemented formation-control model.

The investigation was driven by falsification. A deterministic communication-error bound was analytically valid but produced a Pareto frontier dominated by periodic communication. An online centralized current-state oracle, using the exact action cross term but no future realizations, exposed substantial scheduling headroom in some nonstationary regimes and none in a congestion boundary. Explicit receiver feedback

For every author: [department, institution, complete mailing address, telephone number, email address, and ORCID required].

Corresponding author and preferred proof address: [VERIFIED NAME AND CONTACT REQUIRED].

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often consumed that headroom. Finally, an exact ACK-free receiver-memory belief was well calibrated but could not identify the baseline component of the action cross term. These findings motivated an information-limit result rather than another trigger.

The contributions are:

1) We derive the exact finite-horizon marginal communication-action value as an affine functional of an augmented state for a fixed action response.
2) We give the absolute and relative action-value row-space conditions, crediting them as functional-observability specializations rather than a new observability primitive.
3) We give an exact set-valued fiberwise action-decision criterion and a closed-form local decision margin. Exact value reconstruction is sufficient but not necessary for an action decision.
4) On a genuinely threshold-straddling binary fiber, we quantify the irreducible value radius and deterministic and randomized local minimax loss.
5) We derive the absolute and relative supplementary linear-statistic dimensions and prove that ranking discards at most one dimension, exactly when the hidden value image contains a common offset. Coalition ownership and acquisition cost are treated separately.
6) For the sampled formation architecture, a frozen 72-cell registry gives 1320/1320 unit-response functional non-identifiability results. R2.6 identifies one common-offset reduction in every sender-cell, while ten frozen N=5 actual actions retain independently replayed reachable binary ambiguity.

The results are scoped to exact affine action-value identifiability in the stated sampled linear model. They do not assert that distributed agents can never make useful stochastic decisions, that feedback is always uneconomic, or that adaptive communication is universally inferior to periodic service.

## II. Related Work and Positioning

### A. From event conditions to stale receiver memories

Event-triggered control replaces periodic sampling by state-dependent conditions while retaining stability or performance guarantees [1]–[3], [12], [13]. Loss and delay make the sender and receiver disagree about the control information in use. Decentralized double-integrator consensus under unreliable links is treated in [14]; ACK and no-ACK event-triggered designs under packet loss appear in [15]; and model-based triggering without ACK appears in [16]. Most closely, Viel et al. maintain packet-loss hypotheses for neighbor-held states in distributed formation control [17]. Thus no-ACK receiver memory inference is established prior art. Our issue is what remains hidden after that receiver-memory uncertainty is represented exactly.

### B. Freshness and control value

AoI measures time since generation of the freshest received update [4]. Goal-oriented variants such as age of incorrect information refine freshness by state relevance [5], [18]; a recent multi-agent overview organizes goal-oriented communication by the task information being exchanged [19]. These metrics are useful network summaries but are not generally sufficient statistics for the finite-horizon marginal value of an action. VoI-based feedback control makes that marginal value explicit and can yield optimal threshold structures under its prescribed information model [6]. Recent finite-horizon scheduling under packet drops likewise uses ACK-derived error covariance as a fully observed Markov state [20]. Formation scheduling has also combined value proxies, graph relevance, and resource allocation [21]. We neither introduce the VoI concept nor optimize a scheduler. Bayesian and VoI policies may act optimally under their stated prior and information model without observing a realized clairvoyant value. We instead separate exact fixed-response value reconstruction from reproducing a full-information optimal action on a declared fiber.

### C. Information structures and functional observability

Who knows what and when is central to decentralized control [7], [8]. Common-information methods can convert specific partial-history-sharing problems into coordinated stochastic control problems [9], but they require a declared probability model and information sharing architecture. Multi-agent return-gap communication clusters local observations using joint action-value vectors and bounds expected loss relative to full observability [22]. That learned stochastic-code problem differs from exact affine reconstruction, local set-membership regret, and instantaneous linear-statistic dimension. Functional observability asks whether selected linear functionals, rather than the full state, can be reconstructed [10]. Its row-space characterizations and target-estimation questions are established, including for large networks [11]. Recent work provides modal and structural characterizations and optimal sensor-placement results [23], sample-based conditions for intermittent or irregular measurements [24], and reduced-order functional-observer existence and pole-placement methods [25]. Structural refinements cover generically diagonalizable systems [26] and fixed-observable-set characterizations [27]; target control/observation duality is developed in [28].

These results imply three explicit boundaries. Our current- and finite-history row-space conditions are applications of functional and sample-based functional observability, not new generic tests. Our coalition search is an instantiated functional sensor-selection problem, not a new optimization class. Finally, the instantaneous missing-statistic dimension derived here is neither minimum sensor count nor functional-observer order. Our current-state test is specifically an instantaneous functional reconstructibility application; it is not a functional-observer existence, detectability, order, or pole-placement result. Our addition is the communication-decision synthesis: a conditional fixed-response value, the absolute/relative/fiberwise-decision ladder, local binary minimax regret, the common-offset rank gap, and the separation between linear-statistic dimension and network acquisition cost.

### D. Control–communication co-design

Data-rate theorems and networked-control studies quantify how communication constrains stability and performance [29]–[31]. Modern surveys emphasize that information has several operational meanings in estimation and control [32]. Feedback itself consumes network opportunities and can reverse freshness conclusions [33]. Our break-even analysis is deliberately modest: it asks whether the measured performance headroom can pay for the messages needed to expose a hidden value statistic under a frozen packet-frequency accounting model.

## III. Sampled Distributed Formation and Communication Model

### A. Formation dynamics

Agent 1 is an analytical leader and $\mathcal{F} = \{2, \ldots, N\}$ are followers. At sample $k$, follower i has position $p_i \in \mathbb{R}^3$, velocity $v_i \in \mathbb{R}^3$, and input $u_i$. The implemented semi-implicit double-integrator update is

$$v_i(k+1) = v_i(k) + hu_i(k), \tag{1}$$

$$p_i(k+1) = p_i(k) + hv_i(k) + h^2 u_i(k). \tag{2}$$

For desired offset $r_i$, define

$$e_i = p_i - p_1 - r_i, \qquad w_i = v_i - v_1. \tag{3}$$

The directed convention is $A_{ij} = 1$ when controller i uses a packet sent by j. The evaluated graph is fixed and its follower-grounded dynamics are Schur under the declared gains. To make the implementation instance explicit,

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad \pi = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \end{bmatrix}^\top, \tag{4}$$

with $s_i = 1$, $K_p = 1.8$, $K_v = 2.2$, $K_{pL} = 1.5$, and $K_{vL} = 1.8$.

Receiver i stores the newest accepted ordinary neighbor payload $\hat{x}_{ij} = (\hat{p}_{ij}, \hat{v}_{ij})$. Pinned followers separately store $(\hat{p}^L_{i1}, \hat{v}^L_{i1}, \hat{a}^L_{i1})$. Newer-generation acceptance prevents an obsolete reordered packet from replacing newer memory. DATA may be lost or delayed; an ACK architecture may separately report acceptance, but the information-limit theorem only uses the variables actually supplied to the sender.

Before acceleration saturation, the implemented control law is

$$\begin{aligned} u_i = &- K_p s_i \sum_j A_{ij}\left[(p_i - \hat{p}_{ij}) - (r_i - r_j)\right] \\ &- K_v s_i \sum_j A_{ij}(v_i - \hat{v}_{ij}) - \pi_i K_{pL}[(p_i - \hat{p}^L_{i1}) - r_i] \\ &- \pi_i K_{vL}(v_i - \hat{v}^L_{i1}) + \pi_i \hat{a}^L_{i1}. \end{aligned} \tag{5}$$

Its exact-information part is $u^\star_F = -H_p e - H_v w + \Pi \mathbf{1} a_1$; stale memories add a command residual $d^c$. With $y = \mathrm{col}(e, hw)$, the exact unsaturated sampled update can be written

$$y_{k+1} = A_h y_k + D_h \,\mathrm{col}(\chi_k, \upsilon_k), \tag{6}$$

where $\chi_k$ is the analytical-leader position-step residual and $\upsilon_k = h^2 d^c_k + h^2 \Pi \mathbf{1} a_1(k) - h\mathbf{1}[v_1(k+1) - v_1(k)]$. This form

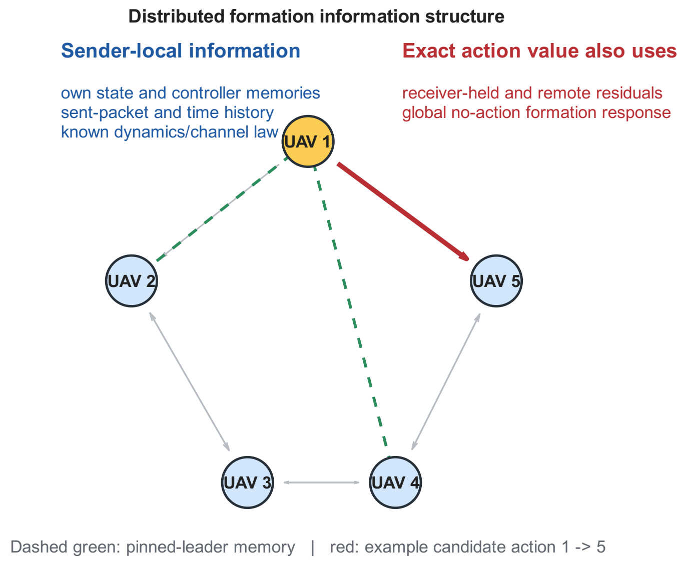


Fig. 1. Evaluated distributed information architecture. The exact value of the highlighted candidate depends on the global no-action response, not only on the sender's own state and estimate of receiver memory.

follows from the actual velocity-first integrator; it is not a continuous-time approximation.

### B. Information available at a sender

At a decision by source j, the strongest deterministic local information used in this paper contains its exact own position and velocity; every ordinary and pinned receiver memory feeding its own controller; its exact unsaturated command; graph, gains, offsets, clock, and channel law; and its own sent-DATA history. The leader additionally knows its reference position, velocity, and acceleration. It does not include state stored at another receiver or remote physical state that has not been communicated. We express the state-dependent part as

$$o_j = C_j \xi + d_j. \tag{7}$$

Known packet histories or action labels change affine constants but do not add hidden-state rows. Figure 1 illustrates the distinction.

## IV. Finite-Horizon Value of a Communication Action

### A. Generic linear/affine closed loop

Consider a finite-dimensional affine closed loop

$$x_{k+1} = Ax_k + Bu_k + d. \tag{8}$$

For a fixed decision point, collect the no-action performance output over a finite horizon as

$$z = Fx + r. \tag{9}$$

A candidate communication action a induces a fixed response $g_a$ in the same stacked output. With symmetric $Q \succeq 0$, define benefit as no-action cost minus action cost:

$$\begin{aligned} q_a(x) &= \|Fx + r\|^2_Q - \|Fx + r + g_a\|^2_Q \\ &= \ell_a^\top x + c_a, \end{aligned} \tag{10}$$

TABLE I
CLOSEST RESEARCH THREADS AND THE DISTINCTION OF THIS PAPER.

| Thread | Established capability | Remaining question | Position of this paper |
|---|---|---|---|
| Event-triggered and lossy formation | Stability/performance triggers under distributed state error, loss, or delay | Is exact marginal action value available to the transmitter? | No new trigger; a valid bound-trigger counterexample motivates the information question. |
| AoI / goal-oriented freshness | Timeliness and incorrectness metrics | Does freshness determine the closed-loop action cross term? | AoI is not treated as a sufficient control-value statistic. |
| VoI scheduling | Marginal control benefit and threshold scheduling under specified information | Is the value statistic itself identifiable under a distributed sender map? | Studies this precondition; makes no optimal-scheduler claim. |
| Decentralized information structures | Formal dependence of decisions on distributed histories | Which exact functional is missing for communication decisions? | Instantiates the information question with finite-horizon communication-action value. |
| Functional and sample-based observability | Row-space/modal/structural tests, sampling conditions, observer design, and target-sensor selection | What irreducible value and decision loss follows when a communication-action functional is hidden? | Uses established tests; adds action-specific minimax regret, simultaneous-action dimension, acquisition separation, and formation closure. |

where

$$\ell_a = -2F^\top Q g_a, \qquad c_a = -2r^\top Q g_a - g_a^\top Q g_a. \tag{11}$$

Positive normalizations, including a delivery probability, do not change identifiability or sign. If the response depends on an unknown state, that state must be augmented or the response must be conditioned on a hypothesis; otherwise (10) does not apply. This quantity is a conditional, fixed-response, finite-horizon marginal output-cost benefit. It freezes the decision point, no-action baseline, controller, horizon, and candidate response; it is not a Bellman value, state-action $Q$-function, or general stochastic VoI that reoptimizes future controls, schedules, or channel outcomes.

### B. Formation action response

For the formation case, the augmented state contains exactly the variables needed to construct the frozen no-action response: follower formation error and scaled relative velocity, analytical-leader state, eight ordinary receiver memories, and two pinned-leader memories. For one Cartesian axis,

$$\begin{aligned}\xi^a = \mathrm{col}\big(&e^a, hw^a, p_L^a, hv_L^a, h^2 a_L^a,\\ &\{\hat{p}_{ij}^a, h\hat{v}_{ij}^a\},\\ &\{\hat{p}_{iL}^a, h\hat{v}_{iL}^a, h^2\hat{a}_{iL}^a\}\big).\end{aligned} \tag{12}$$

The one-axis dimension is 33 and $\xi = \mathrm{col}(\xi^x, \xi^y, \xi^z) \in \mathbb{R}^{99}$.

Recursion of (6) over H samples yields implementation-derived matrices, not fitted matrices,

$$\mathrm{vec}(Z_k) = F_H \xi_k + r_H. \tag{13}$$

For link $j \longrightarrow i$, receiver-memory hypothesis s, successful-delivery probability $p_s$, $m = N-1$, and fixed output response $g_{ij,s}$,

$$q_{ij,s} = -\frac{2p_s}{m} g_{ij,s}^\top (F_H \xi_k + r_H) - \frac{p_s}{m} \|g_{ij,s}\|_2^2. \tag{14}$$

Here $Z_k$ stacks H sampled follower-position errors. Consequently $q_{ij,s}$ is a per-follower sum of squared sampled position-error differences, measured in $\mathrm{m}^2$; it is not an RMSE or time-integrated mission cost. For a receiver-memory belief $b(s)$, the exact conditional expression is $\bar{q}_{ij} = \sum_s b(s) q_{ij,s}$. The coefficient and constant are averaged across hypotheses. We do not evaluate the nonlinear quadratic at an expected receiver state. When $\xi_k$ contains components hidden from the sender, this receiver-memory marginalization is not $\mathbb{E}[q \mid \mathcal{I}_j]$ unless an additional joint state prior is specified.

## V. INFORMATION-STRUCTURE IDENTIFIABILITY

### A. Static current-value reconstructibility

The following classical row-space fact is specialized to the conditional fixed-response communication benefit. It is used as a prerequisite, not claimed as a new functional-observability theorem.

**Theorem 1** (Static fixed-response action-value identifiability)**.** *Let $q_a(x) = \ell_a^\top x + c_a$ and $o = Cx + d_o$. On $\mathbb{R}^n$, or locally on a feasible set containing an open relative neighborhood along* $\ker C$*, the exact action value is a function of $o$ if and only if*

$$\ell_a \in \mathrm{row}(C). \tag{15}$$

*Proof.* If $\ell_a = C^\top \alpha$, then $q_a = \alpha^\top (o - d_o) + c_a$. Conversely, any $v \in \ker C$ leaves o unchanged. Identifiability requires $q_a(x+v) - q_a(x) = \ell_a^\top v = 0$ for every such v. Hence $\ell_a \in (\ker C)^\perp = \mathrm{row}(C)$. □

For $p$ actions, stack $\ell_a^\top$ in $L \in \mathbb{R}^{p \times n}$. Let $B \in \mathbb{R}^{(p-1)\times p}$ have full row rank and $\ker B = \mathrm{span}\{\mathbf{1}_p\}$; a reference-action difference basis is one choice. Then $L_{\mathrm{rel}} = BL$ contains a basis of pairwise value coefficients.

**Proposition 1** (Relative action-value identifiability)**.** *Under the same local-richness condition as Theorem 1, all pairwise action-value differences are identifiable from $o$ if and only if*

$$BL(I - C^\dagger C) = 0. \tag{16}$$

*This condition is independent of the chosen full-rank difference basis.*

*Proof.* The vectors $e_a - e_b$ span $\mathbf{1}_p^\perp = \mathrm{row}(B)$. Thus every difference coefficient is in $\mathrm{row}(C)$ exactly when every row of $BL$ is, which is equivalent to (16). Two admissible bases differ by an invertible left factor and hence generate the same row space after multiplication by $L$. □

Absolute-value identifiability implies relative-value identifiability, which is sufficient to recover the complete ranking and set-valued argmax. Neither converse holds: an unobserved offset common to all values cancels from $BL$, and a bounded information fiber can have a common optimal action even while a pairwise value varies on that fiber. These are action-value applications of row-space geometry, not a new functional-observability primitive.

The normalized diagnostic used later is

$$\delta_C(\ell) = \frac{\|(I - C^\dagger C)\ell\|_2}{\max(\|\ell\|_2, \epsilon)}. \tag{17}$$

A positive numerical residual applies the theorem to a given matrix; it is not itself the theorem.

### B. Finite local history

Suppose that during a declared interval the augmented dynamics and measurements are

$$x_{t+1} = A_0 x_t + a_0, \qquad o_t = C x_t + d. \tag{18}$$

Define

$$O_L = \mathrm{col}(C, CA_0, \ldots, CA_0^L). \tag{19}$$

**Theorem 2** (Finite-history action-value identifiability)**.** *At time L, an action value with coefficient $\ell_a$ is identifiable from $o_0, \ldots, o_L$ if and only if*

$$(A_0^L)^\top \ell_a \in \mathrm{row}(O_L). \tag{20}$$

*After the observability row space has stabilized, later samples add no observation direction. They cannot recover a decision-time value whenever its propagated coefficient remains outside that stabilized row space.*

*Proof.* The affine solution gives $x_L = A_0^L x_0 + \sum_{t=0}^{L-1} A_0^{L-1-t} a_0$. Thus the coefficient of the time-$L$ value in $x_0$ is $(A_0^L)^\top \ell_a$, while the stacked history's state-dependent part is $O_L x_0$. Theorem 1 gives (20). Cayley–Hamilton implies that the row span generated by $C, CA_0, \ldots$ stabilizes in finite dimension. □

This is the corresponding sampled functional-observability specialization, not a new result based merely on sampling. The last sentence is conditional on unchanged maps and a coefficient that retains its hidden component. A hidden mode annihilated by the dynamics can cease to affect a later value; that is loss of dependence, not recovery of information. A delivered packet, new sensor, or shared statistic changes the information structure.

### C. Fiberwise optimal actions and binary regret

Let $\mathcal{X}$ be the declared operating set and

$$\mathcal{X}(o) = \{x \in \mathcal{X} : Cx + d_o = o\}, \qquad \mathcal{A}^*(x) = \arg\max_{a \in \mathcal{A}} q_a(x). \tag{21}$$

**Theorem 3** (Exact fiberwise action decidability)**.** *A deterministic sender-local action can be full-information optimal for every state compatible with o if and only if*

$$\bigcap_{x \in \mathcal{X}(o)} \mathcal{A}^*(x) \neq \emptyset. \tag{22}$$

*Proof.* All states in the fiber produce the same $o$, so a deterministic local rule must choose one action $a_o$. It is optimal for every compatible state exactly when $a_o \in \mathcal{A}^*(x)$ for every $x$, which is (22). □

This criterion preserves ties: any action in the intersection is valid. A particular single-valued tie-break would additionally have to be constant on the fiber and is not imposed here.

For the Euclidean local fiber $\mathcal{X}(o) = \{x_0 + v : v \in \ker C, \ \|v\|_2 \le \rho\}$, let $P_\perp = I - C^\dagger C$, $d_{ab} = q_a - q_b$, and $h_{ab} = P_\perp(\ell_a - \ell_b)$.

**Proposition 2** (Euclidean-fiber decision margin)**.** *For every action pair,*

$$d_{ab}(\mathcal{X}(o)) = [d_{ab}(x_0) - \rho\|h_{ab}\|_2, d_{ab}(x_0) + \rho\|h_{ab}\|_2]. \tag{23}$$

*Action a is common-optimal if and only if, for every $b \neq a$,*

$$m_{ab} := d_{ab}(x_0) - \rho\|h_{ab}\|_2 \ge 0. \tag{24}$$

*Strict inequalities make a uniquely optimal throughout the fiber.*

*Proof.* For $v \in \ker C$, $d_{ab}(x_0 + v) = d_{ab}(x_0) + h_{ab}^\top v$. Cauchy–Schwarz gives the endpoints, attained by $v = \pm\rho h_{ab}/\|h_{ab}\|_2$ when $h_{ab} \neq 0$; the interval is a singleton otherwise. Action $a$ is optimal throughout exactly when the minimum of every difference against $b \neq a$ is nonnegative. □

The scalar margin $m_a = \min_{b \neq a} m_{ab}$ is positive for strict fiberwide optimality, zero at a tie boundary, and negative when optimality is not guaranteed. Figure 5 shows the corresponding hidden direction. This margin does not introduce a noisy-observation model.

Transmit/no-transmit is the two-action case. With $q_0 = 0$ and net transmit benefit $q = q_1 - \tau$, let its compatible image be the nonempty closed finite interval $Q(o) = [q_-, q_+]$. A nonnegative lower endpoint makes transmit common-optimal; a nonpositive upper endpoint makes no-transmit common-optimal; strict zero straddling leaves no common action. Define

$$q_c = \frac{q_+ + q_-}{2}, \qquad \Delta_q = \frac{q_+ - q_-}{2}. \tag{25}$$

**Theorem 4** (Binary value radius and ambiguous-fiber regret)**.** *The minimax error of a sender-local point estimate is*

$$\inf_{\widehat{q}(o)} \sup_{x \in \mathcal{X}(o)} |\widehat{q}(o) - q(x)| = \Delta_q, \tag{26}$$

*attained by $\widehat{q} = q_c$. If $q_- < 0 < q_+$, a randomized rule that transmits with probability $p$ has endpoint regrets $R_+(p) = (1-p)q_+$ and $R_-(p) = p(-q_-)$, with*

$$p^* = \frac{q_+}{q_+ + (-q_-)}, \quad R^*_{\mathrm{rand}} = \frac{q_+(-q_-)}{q_+ + (-q_-)} > 0, \tag{27}$$

*while deterministic rules incur*

$$R^*_{\mathrm{det}} = \min\{q_+, -q_-\} > 0. \tag{28}$$

*Proof.* The inequality $q_+ - q_- \le |q_+ - y| + |y - q_-|$ lower-bounds the worst endpoint error by $\Delta_q$, attained by the midpoint. Under strict straddling, Theorem 3 rules out a common deterministic action. Positive-endpoint regret decreases with $p$, while negative-endpoint regret increases; equalizing them proves (27). The choices $p = 0, 1$ give (28). □

If $q_- = 0 < q_+$, transmit is common-optimal; if $q_- < 0 = q_+$, no-transmit is common-optimal; and if both vanish, both are. On $[-a, a]$, $p^* = 1/2$, $R^*_{\mathrm{rand}} = a/2$, and $R^*_{\mathrm{det}} = a$. These are local one-decision minimax results, not Bayesian, dynamic-regret, or universal stochastic-control bounds.

## VI. Minimum Supplementary Information and Coalition Structure

### A. One or multiple candidate actions

Using the action matrix and difference basis from Proposition 1, define

$$M = LP_{\perp}, \qquad M_{\text{rel}} = BLP_{\perp}. \tag{29}$$

**Theorem 5** (Absolute and relative linear-statistic dimensions)**.** *Under an instantaneous, noiseless, arbitrary-precision real-valued linear supplementary map $Hx$, the minimum scalar dimensions for recovering all absolute values and all relative values are, respectively,*

$$r^*_{\text{value}} = \operatorname{rank}(M), \qquad r^*_{\text{rel}} = \operatorname{rank}(M_{\text{rel}}). \tag{30}$$

*Moreover,*

$$r^*_{\text{value}} - r^*_{\text{rel}} = \dim\left(\operatorname{im}(M) \cap \operatorname{span}\{\mathbf{1}_p\}\right), \tag{31}$$

*so $r^*_{\text{value}} - 1 \le r^*_{\text{rel}} \le r^*_{\text{value}}$.*

*Proof.* For either target matrix $T = L$ or $T = BL$, identifiability from $\operatorname{col}(C, H)x$ requires the rows of $HP_{\perp}$ to span those of $TP_{\perp}$, giving the rank lower bound. Choosing $H$ as a row basis of $TP_{\perp}$ attains it because $T = TP_C + TP_{\perp}$. For the gap, restrict the linear map $B$ to $\operatorname{im}(M)$. Its image is $\operatorname{im}(BM)$ and its kernel is $\operatorname{im}(M) \cap \ker B$. Rank–nullity and $\ker B = \operatorname{span}\{\mathbf{1}_p\}$ give (31); the intersection has dimension zero or one. □

Strict reduction occurs exactly when $\mathbf{1}_p \in \operatorname{im}(M)$: some hidden perturbation changes every action value by the same nonzero amount. Indeed, $Mz = \alpha\mathbf{1}_p$, $\alpha \neq 0$, is equivalent to the hidden perturbation $v = P_{\perp}z \in \ker C$ satisfying $Lv = \alpha\mathbf{1}_p$. Exactly-one-of-$p$ ranking discards this common hidden offset. The quantity $r^*_{\text{rel}}$ is the minimum linear-statistic dimension for recovering all relative action values, not the minimum information for argmax; Theorem 3 can certify a fiber-specific winner with less.

**Corollary 1** (At-most-one action family)**.** *Augment the communication actions by the genuine no-transmission action $q_0 = 0$. Then the minimum relative-value dimension of the augmented family is $r^*_{\text{rel,aug}} = r^*_{\text{value}}$.*

*Proof.* The augmented hidden-value matrix is $\widetilde{M} = [0; M]$. Its image has zero first coordinate, hence has trivial intersection with $\operatorname{span}\{\mathbf{1}_{p+1}\}$. Equation (31) then gives no relative rank loss, while $\operatorname{rank}(\widetilde{M}) = \operatorname{rank}(M)$. □

*Scheduling semantics:* Exactly-one-of-$p$ communication-action ranking may discard one common hidden offset. At-most-one transmission includes the genuine no-transmission action $q_0 = 0$, so action-to-action cancellation does not make transmit versus no-transmit locally decidable. Fiber-specific dominance may still make a particular action decidable and is tested by Theorem 3. For a single nonidentifiable value, one extra scalar $\ell_{\perp}^{\top}x$ suffices. All these counts are linear-statistic dimensions, not packets, bits, rates, observer order, or distributed acquisition cost.

### B. Information ownership

Let agent $i$'s strongest map be $C_i$. For coalition $\mathcal{S}$, define $C_{\mathcal{S}} = \operatorname{col}_{i\in\mathcal{S}} C_i$.

**Corollary 2** (Coalition identifiability)**.** *Coalition $\mathcal{S}$ identifies a fixed action value if and only if*

$$\ell \in \operatorname{row}(C_{\mathcal{S}}). \tag{32}$$

*A minimum owning coalition solves*

$$\min_{\mathcal{S}} |\mathcal{S}| \quad \textit{subject to} \quad \ell \in \operatorname{row}(C_{\mathcal{S}}). \tag{33}$$

Corollary 2 is closely related to functional sensor selection. We exhaustively enumerate only the finite evaluated registry through N=9 and claim no new generic combinatorial algorithm. Even when a coalition owns the required rows, a causal aggregation protocol must still communicate them at a nonzero frequency, delay, and reliability cost.

## VII. Communication Cost of Acquiring Value Information

Let $C_P(E)$ be the cost of the frozen periodic Pareto frontier interpolated at formation error $E$. Suppose a policy informed by supplementary statistics uses DATA cost $D$ and information streams $t$ with rate $R_t$ and normalized price $\eta_t$. A necessary accounting condition for performance-matched Pareto benefit is

$$D + \sum_t \eta_t R_t < C_P(E). \tag{34}$$

For a single feedback stream with rate $A$,

$$\eta^\star = \frac{C_P(E) - D}{A} \tag{35}$$

is its break-even price. Negative $\eta^\star$ means DATA alone is already more costly than periodic at matched error. At feedback price $\eta_f$, the affordable feedback fraction is $f^\star = \eta^\star/\eta_f$, clipped to $[0, 1]$ for operational interpretation.

Equation (34) is not an optimality theorem. It prevents a common category error: a one-dimensional sufficient statistic is not free. Under a packet-frequency metric, piggybacking adds no packet only if an existing packet is sent anyway; bytes, airtime, collision risk, and joules remain outside that metric.

## VIII. Formation-Control Case Study

### A. Implementation-derived matrices and cross-instance validation

The frozen case uses N=5, $h = 0.02$ s, four followers, eight ordinary controller links, two pinned-leader streams, H=25, and D=4 samples. Its augmented state has 33 entries per axis and 99 in 3-D. The matrices $F_H$, action responses, sender maps, and held-memory dynamics are assembled directly from the implemented graph, gains, controller, and integrator. For each controller-relevant payload, a fixed unit command correction $[1, 0, 0]^{\top}$ and fixed receiver-memory hypothesis define $g$. The maximum discrepancy between the affine representation and the centralized calculation is $4.34 \times 10^{-16}$.

To test whether the information-structure observation is peculiar to that instance, we froze a new Cartesian registry

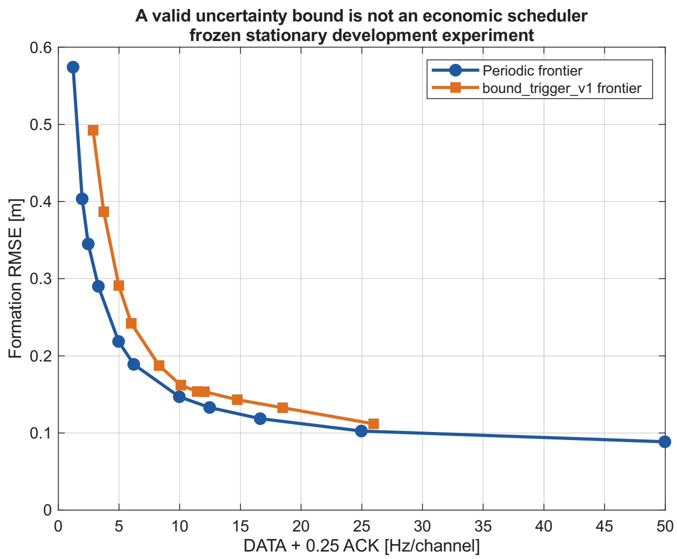


Fig. 2. Frozen stationary frontiers from five paired development seeds. The analytically valid bound-based mechanism is dominated by the periodic frontier over their shared domain. The comparison is descriptive, not a population claim.

before viewing its results: $N \in \{5, 7, 9\}$, three deterministic connected graphs, even/odd follower pinning, $H \in \{15, 25\}$, and $D \in \{2, 4\}$. All 72 configurations satisfy the same symmetric grounded Schur scope. Fixed-payload histories use the complete finite-dimensional horizon $n_{\text{free}} - 1$. Before grounding, ring2 is a cycle, sparse4 adds cyclic distance-two edges, and geometric is the first connected radius graph on the deterministic lattice in the sequence 0.60, 0.75, . . . m. Table II summarizes all 1320 action rows.

At $10^{-10}$, compressed-null and independent direct-row-space calculations agree for 1320/1320 rows; the normalized hidden residual spans $3.03 \times 10^{-6}$–0.99999. Across the tolerance grid, raw rank is stable for only 608/1320 rows and raw $\operatorname{rank}([C; \ell^{\top}])$ adds one in 1222 rows at $10^{-6}$ and 614 at $10^{-14}$. An independent reconstruction matches all 6600 stored decisions, while normalized-basis augmentation gives 1320/1320 at every cutoff. Selected 80-digit checks, including each cell's minimum, remain positive. The claim is therefore scale-controlled functional membership in a finite registry, not invariant raw-rank bookkeeping or an arbitrary-graph/gain theorem.

The fixed unit-response correction fixes $g$ and must not be confused with a trajectory-realized scheduling action. Zero response has zero value, and special nonzero responses can cancel a hidden projection. The next subsection therefore audits actual state-generated corrections separately.

### B. Reachable value-sign witnesses

Affine nonidentifiability alone could be dismissed as an inadmissible state perturbation. We therefore audit each of the ten controller-relevant action types in the frozen N=5 catalog. Receiver memories are initialized consistently and held without a DATA update until the decision. A deterministic source excitation creates an actual 0.01 m/s$^2$ x-axis controller correction. For follower senders, the larger of the source's position and scaled-velocity transfers is used; leader actions use constant velocity. We then project the exact fixed-response value gradient into the consistent-initialization subspace that simultaneously preserves the complete sender history, target receiver memory, and actual action response. Symmetric perturbations about the exact affine zero crossing are replayed through the implemented controller and integrator.

For each of the ten catalog action types, an actual state-generated nonzero response yields an admissible opposite-sign pair; six have follower senders not covered by the earlier representative construction. Independent replay from persisted coordinates recomputes the physical controller and centralized trajectories rather than reusing the affine sign expression. Correction, payload, branch, and response match within every pair. Maximum history, dynamic, and affine-versus-oracle residuals are $7.98 \times 10^{-20}$, $4.17 \times 10^{-16}$, and $4.07 \times 10^{-20}$; response mismatch is zero. The $10^{-4}$ coordinate half-separation yields a smallest endpoint $8.52 \times 10^{-8}$, over $8.5 \times 10^{4}$ sign tolerances, and minimum saturation margin 0.952 m/s$^2$. Values in Table III are in m$^2$ under the sampled per-follower normalization following (14).

All ten intervals strictly straddle zero, so Theorem 3 leaves no common deterministic transmit/no-transmit action. Because the endpoints are centered symmetrically, Theorem 4 gives $p^* = 1/2$, randomized minimax regret $\Delta_q/2$, and deterministic minimax regret $\Delta_q$ for every row. These are local one-decision losses, not expected mission-level bounds. Practical scale is heterogeneous: the normalized radius exceeds 0.15 for both pin and both leader ordinary actions, but is below $3.5 \times 10^{-4}$ for two follower actions. Seven of ten actions cross zero somewhere on the selected 32-state grid; a constructed reachable witness exists for each catalog type. The maximum is 0.232, or 0.145 and 0.175 under RMS and IQR/1.349 scaling; the two smallest remain order $10^{-4}$. Figure 6 shows this heterogeneity.

### C. Statistic dimension versus coalition size

For every fixed payload, $\operatorname{rank}(M) = 1$: one linear scalar closes its individual algebraic deficit. However, no receiver alone and no sender–receiver pair identifies that scalar. Exhaustive enumeration of all subsets of the five implemented agent maps finds that every sender needs four additional agents; the only minimum coalition contains all five UAVs. This is conditioned on the evaluated graph, H=25, the selected global stacked formation-error output with identity weighting, and the declared current information maps. A localized or decomposable objective can yield a smaller coalition. It is not a claim that all formations require global information.

The fixed-action statement underuses Theorem 5. We therefore stack every controller-relevant current action of each actual sender in $L_j$, using a common full 99-dimensional state and the fixed unit command response. Table IV reports both absolute and relative missing ranks. Sender 1 has four actions with $(r^*_{\text{value}}, r^*_{\text{rel}}) = (3, 2)$; senders 3 and 4 have $(2, 1)$. The $p = 1$ relative ranks are vacuous rankings, not evidence that a meaningful choice is identifiable.

Although $r^*_{\text{value},1} = 3 < 4$, the minimum coalition for its entire missing subspace still contains all five agents. The same

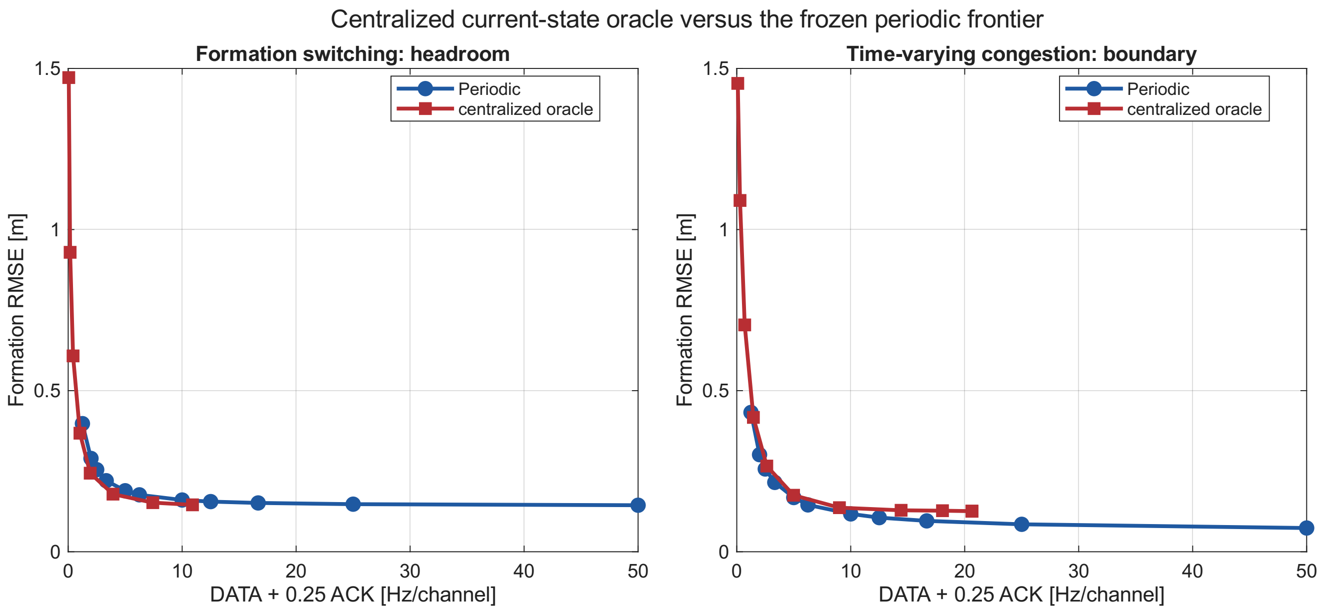


Fig. 3. Frozen five-seed development diagnostic with centralized current plant and receiver information. The conditional fixed-response value creates Pareto headroom in S2 but not in the S4 congestion boundary. The oracle is privileged and nondeployable, and the comparison is non-inferential.

TABLE II

Cross-instance unit-response functional audit. "Rows" aggregates two pinning patterns, two H values, and two D values; every row is nonidentifiable, stable in membership over relative tolerances $10^{-6}$–$10^{-14}$. After the retained row space is orthonormalized and the value row is normalized, every value row adds one dimension. "Raw stable" counts rows whose unnormalized information-map rank itself does not change over that grid.

| Graph | N | Rows/non-ID | Raw stable | Leader $(p_1, r^*_{\text{value}}, r^*_{\text{rel}})$ | $r^*_{\text{rel}}/r^*_{\text{value}}$ | Minimum coalition pattern |
|---|---|---|---|---|---|---|
| geometric | 5 | 48/48 | 48 | (6,4,3) | 0.750 | leader: 5 |
| ring2 | 5 | 80/80 | 56 | (4,3,2) | 0.667 | every sender: 5 |
| sparse4 | 5 | 144/144 | 96 | (6,4,3) | 0.750 | every sender: 5 |
| geometric | 7 | 104/104 | 80 | (7,5,4) | 0.800 | leader: 7; followers: 5 |
| ring2 | 7 | 120/120 | 80 | (5,4,3) | 0.750 | every sender: 7 |
| sparse4 | 7 | 216/216 | 56 | (7,5,4) | 0.800 | every sender: 7 |
| geometric | 9 | 160/160 | 80 | (8,6,5) | 0.833 | leader: 9; followers: 8 |
| ring2 | 9 | 160/160 | 48 | (6,5,4) | 0.800 | every sender: 9 |
| sparse4 | 9 | 288/288 | 64 | (8,6,5) | 0.833 | every sender: 9 |

coalition size holds for every other sender's full action set. Compression of value directions therefore does not imply local ownership or three packets: Theorem 5 counts independent real-valued statistics only if their operands can be assembled before the decision.

The cross-instance audit qualifies the ownership result. Leader compression of absolute values recurs in every registry cell, while every tested follower has $r^*_{\text{value}} = p_j$. R2.6 post-processing finds $r^*_{\text{rel}} = r^*_{\text{value}} - 1$ and the equivalent common-hidden-offset condition in all 448 sender–cells. All 360 nontrivial sets with $p \geq 2$ therefore have strict one-dimensional reduction, but none is fully relative-identifiable; the 88 single-action cases are excluded from that ordering interpretation. Ring2 and sparse4 require the complete N-agent coalition for every sender. On the geometric graph, however, follower-sender value sets require five of seven agents and eight of nine agents, while the leader still requires all agents. Coalition ownership is therefore topology and action-set dependent; the general claim is the row-space condition, not an all-agent rule. Across 448 sender–cell rows, unique minima 5, 7, 8, and 9 occur 120, 120, 56, and 152 times; these counts do not imply packet or acquisition cost.

These ranks concern reconstruction, not the actual argmax on one bounded fiber. The ten persisted action witnesses below use action-specific conditioned fibers, so they are not combined into a multi-action common-maximizer claim.

### D. *Why the information result matters operationally*

Figure 2 retains the mechanism that failed before this theory was formulated. Its trigger uses a valid staleness-to-control uncertainty bound, yet the complete periodic frontier is better throughout the shared development domain. Across a 101-point budget-matched grid, trigger-minus- periodic RMSE averages +0.03023 m and no point favors the trigger. This is a counterexample to the inference that a safe uncertainty bound is automatically a useful scheduler; it is not a population comparison.

To test whether valuable scheduling opportunities existed at all, a centralized online oracle used exact current plant and receiver memories but no future channel outcomes, disturbances, formation changes, or trajectories. It ranked actions only by (14). Figure 3 shows a positive and a negative

TABLE III

N=5 ACTUAL-ACTION REACHABLE VALUE-SIGN WITNESSES. THE NORMALIZED RADIUS DIVIDES $\Delta_q$ BY THE MEDIAN ABSOLUTE EXACT VALUE ON A DETERMINISTIC 32-STATE LOCAL GRID. "CROSS" IS THE FRACTION OF THOSE STATES WHOSE RADIUS- $10^{-4}$ COMPATIBLE INTERVAL CROSSES ZERO; IT IS DESCRIPTIVE, NOT A POPULATION PROBABILITY.

| Action | $q_-$ | $q_+$ | $\Delta_q$ | Normalized radius | Cross |
|---|---|---|---|---|---|
| ordinary $1 \to 2$ | $-3.475 \times 10^{-6}$ | $3.475 \times 10^{-6}$ | $3.475 \times 10^{-6}$ | 0.1590 | 0.1875 |
| ordinary $1 \to 5$ | $-5.344 \times 10^{-6}$ | $5.344 \times 10^{-6}$ | $5.344 \times 10^{-6}$ | 0.1953 | 0.0625 |
| ordinary $2 \to 3$ | $-5.392 \times 10^{-6}$ | $5.392 \times 10^{-6}$ | $5.392 \times 10^{-6}$ | 0.01010 | 0.09375 |
| ordinary $3 \to 2$ | $-8.519 \times 10^{-8}$ | $8.519 \times 10^{-8}$ | $8.519 \times 10^{-8}$ | 0.0002666 | 0 |
| ordinary $3 \to 4$ | $-3.198 \times 10^{-6}$ | $3.198 \times 10^{-6}$ | $3.198 \times 10^{-6}$ | 0.009173 | 0.0625 |
| ordinary $4 \to 3$ | $-5.223 \times 10^{-6}$ | $5.223 \times 10^{-6}$ | $5.223 \times 10^{-6}$ | 0.01080 | 0 |
| ordinary $4 \to 5$ | $-1.333 \times 10^{-7}$ | $1.333 \times 10^{-7}$ | $1.333 \times 10^{-7}$ | 0.0003459 | 0 |
| ordinary $5 \to 4$ | $-3.396 \times 10^{-6}$ | $3.396 \times 10^{-6}$ | $3.396 \times 10^{-6}$ | 0.01099 | 0.0625 |
| pin $1 \to 2$ | $-3.449 \times 10^{-6}$ | $3.449 \times 10^{-6}$ | $3.449 \times 10^{-6}$ | 0.2315 | 0.2500 |
| pin $1 \to 4$ | $-5.002 \times 10^{-6}$ | $5.002 \times 10^{-6}$ | $5.002 \times 10^{-6}$ | 0.2106 | 0.03125 |

TABLE IV

FROZEN N=5 SENDER-WISE ABSOLUTE/RELATIVE MISSING DIMENSION AND OWNERSHIP.

| Sender j | $p_j$ | $r^*_{\text{value}}$ | $r^*_{\text{rel}}$ | Ratio | Minimum coalition | Interpretation |
|---|---|---|---|---|---|---|
| 1 | 4 | 3 | 2 | 0.667 | $\{1,2,3,4,5\}$ | one common hidden offset removed |
| 2 | 1 | 1 | 0 | – | $\{1,2,3,4,5\}$ | ranking vacuous |
| 3 | 2 | 2 | 1 | 0.500 | $\{1,2,3,4,5\}$ | one relative hidden direction |
| 4 | 2 | 2 | 1 | 0.500 | $\{1,2,3,4,5\}$ | one relative hidden direction |
| 5 | 1 | 1 | 0 | – | $\{1,2,3,4,5\}$ | ranking vacuous |

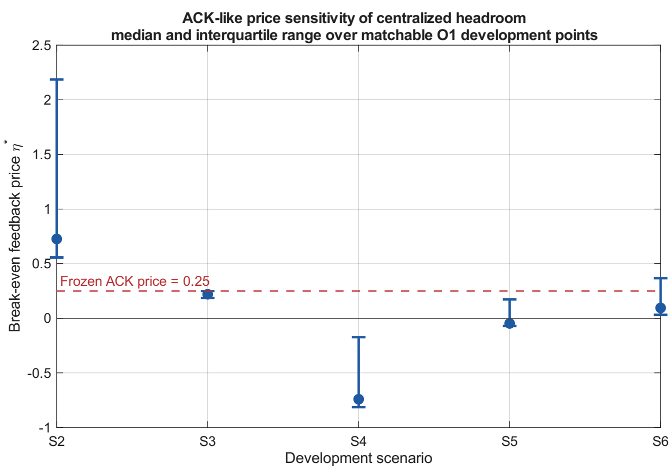


Fig. 4. ACK-like packet-frequency cost sensitivity over frozen, performance-matchable centralized-oracle development points (median and IQR). The dashed line is the preregistered price 0.25. This does not measure the cost of acquiring the all-agent missing statistic.

boundary. In S2 formation switching, oracle minus periodic mean matched RMSE is -0.02381 m and mean matched cost is -1.7088 Hz/channel; the oracle is favored over the complete shared domains. In S4 time-varying congestion, the differences reverse to +0.02595 m and +0.7135 Hz/channel, with zero shared-domain fraction favoring the oracle. The oracle is privileged and is not a proposed algorithm.

Finally, Fig. 4 applies (35) to an ACK-like receiver-information stream. At the frozen ACK price 0.25, S2 can fund one such packet per accepted delivery at all four matchable points, while the median S6 point can fund it on only 37.8% of eligible deliveries. S4 has negative headroom even before feedback. Across all five scenarios, the median break-even prices range from -0.741 to 0.726. This is a necessary cost-

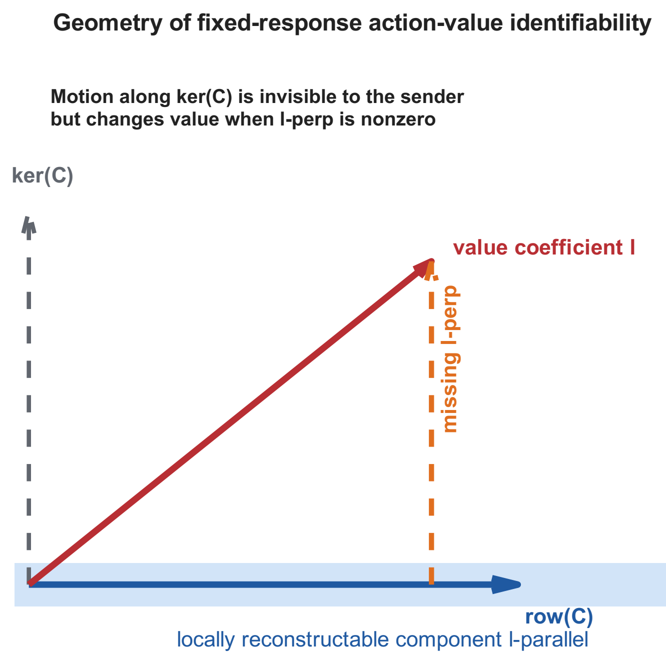


Fig. 5. The available row space determines the reconstructable component of the value coefficient. Motion along the information null space is invisible to the sender but changes value when the missing projection is nonzero.

sensitivity screen, not the measured cost of causally aggregating the all-agent statistic. It explains why algebraic sufficiency and deployable economic value must remain distinct.

## IX. DISCUSSION AND LIMITATIONS

### A. What the result changes

Theorems 1–5 suggest a staged audit for control-aware communication: derive the fixed-response values; distinguish absolute reconstruction, relative reconstruction, and the direct

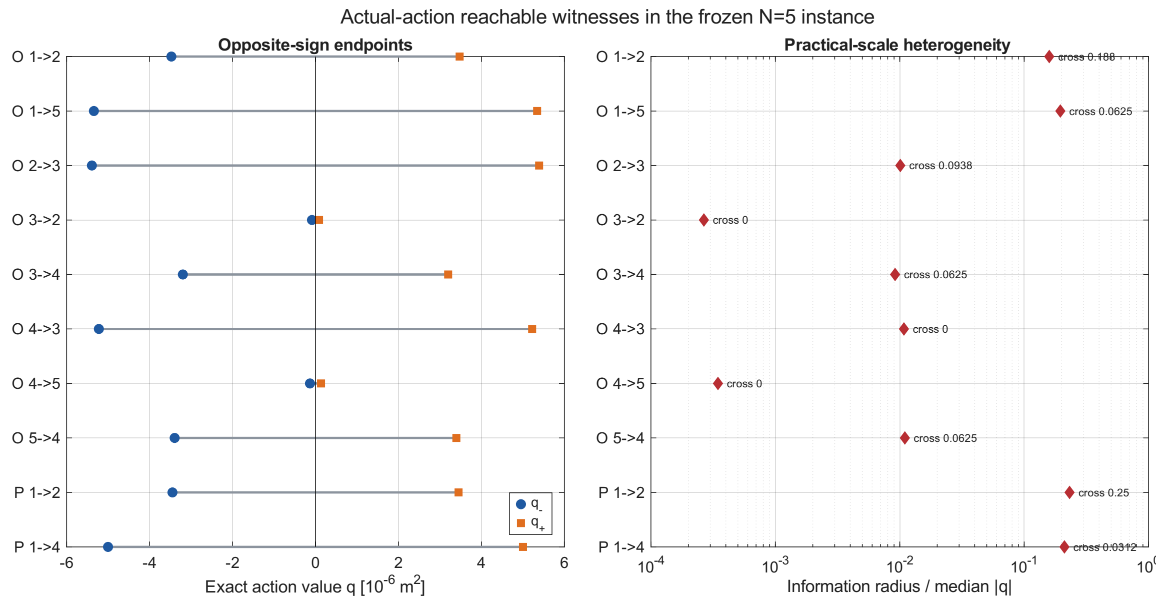


Fig. 6. Implementation-replayed N=5 actual-action witnesses. Left: one constructed actual-response pair for each of ten catalog action types. Right: local information radius normalized by typical exact action-value scale; labels give the deterministic-grid zero-crossing fraction. Small normalized radius for some actions is retained rather than interpreted as uniform practical impact.

common-maximizer test; then separate any missing linear-statistic dimension from the coalition and communication resources needed to construct it. Receiver-memory estimation alone may still miss the projection of the action response onto remote components of the no-action trajectory. Calibrating that memory marginal cannot create measurements of those components.

### B. Boundaries

The analytical case is an unsaturated sampled double integrator with a fixed graph within each decision history, not nonlinear 6-DOF dynamics, saturation, within-history switching, or an interference-limited MAC. The 72-cell audit is a finite registry, not arbitrary-topology, arbitrary-gain, or arbitrary-N theory. Tolerance sweeps and selected 80-digit projections strengthen, but do not replace, a symbolic parameter proof. The minimum residual is only 3.03 times the loosest tolerance, despite a conditional-basis check to $10^{-16}$.

Exact algebraic row-space nonidentifiability is not a robustness or noisy- observation impossibility theorem. Relative-value nonidentifiability likewise does not by itself imply argmax ambiguity; the fiberwise common-maximizer condition is the direct decision test. No multi-action reachable common-fiber experiment was constructed: the ten N=5 witnesses use action-specific fibers.

The value radius and regret are local to a declared compatible set. R2.6 does not solve stochastic or Bayesian scheduling; a specified prior and estimator define a different, untested information model. Randomization reduces worst-case binary regret but does not reveal the hidden sign, and the witnessed normalized ambiguity spans nearly three orders of magnitude.

The frontier uses five paired development seeds as mechanism/counterexample evidence, not held-out superiority or statistical significance. Its DATA + 0.25 ACK packet-frequency axis omits payload length, shared-medium collisions, airtime, and energy; the economics are conditional on that accounting.

Functional-observability and sensor-selection theories are established fields. Our row-space, history, and coalition conditions are action-value specializations, not new generic observability or placement theory. The contribution is their control–communication synthesis with the decision ladder, local margin and loss, rank gap, acquisition cost, and reachable closure.

## X. Conclusion

For a fixed response in a linear sampled loop, finite-horizon quadratic benefit is affine. Absolute reconstruction implies relative reconstruction, but the common-maximizer intersection directly governs local action selection. Its Euclidean-fiber margins are closed form; strict binary threshold straddling has positive local minimax regret. Relative-value recovery can save only the one common-offset dimension, and adding no transmission removes that saving.

All 1320 fixed unit-response functionals in the 72-cell registry violate the declared sender-history row-space condition. All 360 nontrivial action sets save one relative dimension, but none is fully relative-identifiable. Each of ten N=5 action types has a reachable, unsaturated, sender-indistinguishable pair requiring opposite binary decisions; normalized radii range from $2.67 \times 10^{-4}$ to 0.232. The N=5 leader's four actions require three absolute and two relative missing directions, while coalition size varies with topology. These results motivate information-structure and acquisition- cost audits; structured priors, nonlinear theory, and causal acquisition protocols remain future work.

# Appendix A
## Detailed Formation Affine Map

For one axis, let $T_y\xi = y$ and write the exact held-current communication input as $T_\nu\xi + t_\nu$. Initialize $M_0 = T_y$ and $b_0 = 0$. Then recurse as

$$M_r = A_h M_{r-1} + \begin{bmatrix} I \\ I \end{bmatrix} T_\nu, \tag{36}$$

$$b_r = A_h b_{r-1} + \begin{bmatrix} I \\ I \end{bmatrix} t_\nu. \tag{37}$$

With $C_e = [I\ 0]$, stacking $C_e M_r$ and $C_e b_r$ for $r = 1, \ldots, H$ and then for three coordinates yields $F_H$ and $r_H$ in (13). Substitution of a fixed receiver memory removes its coordinates from the free state and moves them into the constant. The action response is generated by the same $A_h$, delay D, gains, and target row used by the centralized diagnostic.

# Appendix B
## Reproducibility Scope

The base theorem validation entry point is `experiments/tcns_information_limits_validation.m`; the R1 depth audit is `experiments/tcns_r1_theory_depth_validation.m`, with unit test `tests/test_tcns_r1_theory_depth.m`. Figure generation is `paper/tcns/manuscript/make_gm_figures.m`. The authoritative base theorem-validation run, generated from source commit `15177a8`, is `results/tcns_information_limits_validation/2026-09-08_095209`. The authoritative R1 rank/regret/multi-action audit, generated from source commit `05fa200`, is `results/tcns_r1_theory_depth_validation/2026-09-08_103325`. The authoritative R2 cross-instance and all-action witness audit, generated from implementation commit `a2cec3b`, is `results/tcns_r2_generalization_validation/2026-09-08_161130`. The independent R2.5 audit is `results/tcns_r2_5_adversarial_validation/2026-09-08_175427`. Registry, witness-scenario, and reference-direction seeds are 27022001, 27020001, and 27022001 plus action index; no stochastic draw enters the witness construction. The R2.6 decision-identifiability post-processing artifact is `results/tcns_r2_6_decision_identifiability/2026-09-08_232116`; its exact source snapshot is included there and committed in `da97e01`, with the rank-gap refinement in `d0edd2c`. It reads only the frozen R2 matrices and witnesses; its test entry point is `tests/test_tcns_r2_6_decision_identifiability.m`. Frozen negative-trigger, centralized-oracle, feedback-accounting, and feedback-economic result directories are named in the repository's claim–evidence ledger. Each result records configuration, seed, source commit, MATLAB release, machine-readable tables, and a console log. No held-out seed or new policy experiment is used in this manuscript.

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